

ME 333
MECHANICS OF MATERIALS

Lecture 14: Bending of Beams I - Review

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Sem I, 2025 - 2026

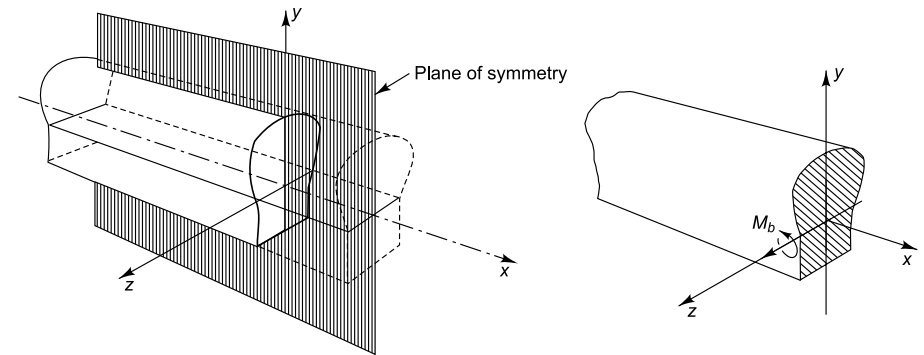
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ME 333

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Pure Bending
Pure Bending

Pure Bending of symmetric beams



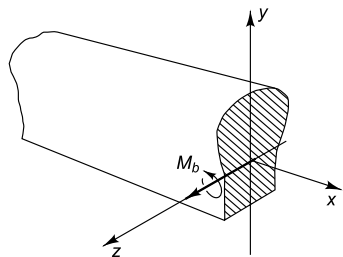
Ref Base Figs: An Introduction to Mechanics of Solids, Crandall et. al.

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Pure Bending
Pure bending of symmetric beams



- originally straight beam
- uniform along its length
- cross-section symmetric about plane of loading
- material properties are constant along the length of the beam

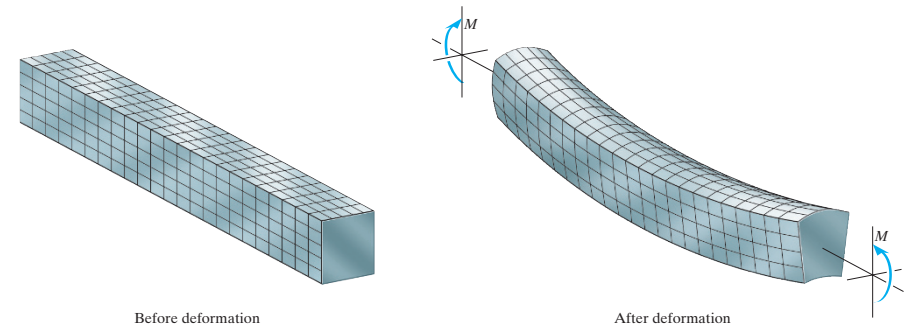
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Ref Base Figs: Statics & Mechanics of Materials, R. C. Hibbeler

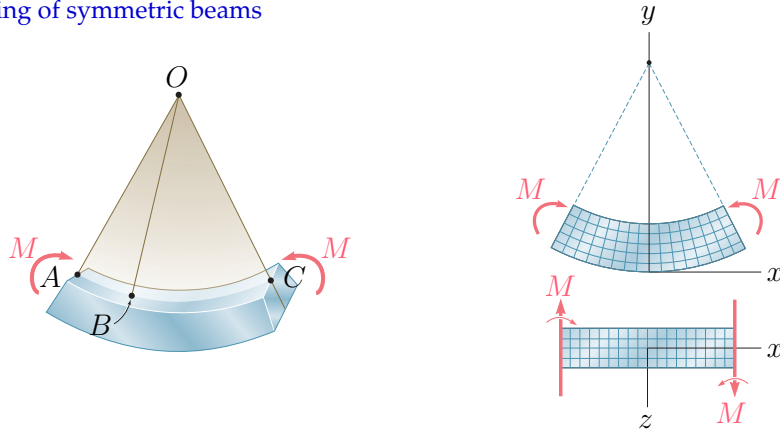
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Pure Bending

Pure Bending of symmetric beams



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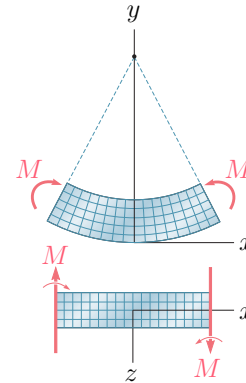
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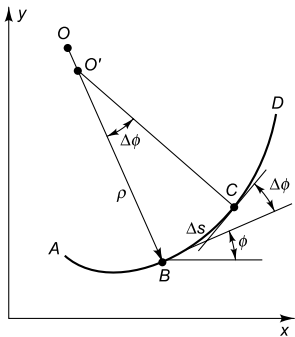
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- straight beam deforms into a curved shape
- horizontal lines become curved
- vertical lines remain straight and rotate - **WHY ?**
- planar cross-sections remain plane post bending - **WHY ?**
- what about deformations in each cross-sectional plane?

Pure Bending

Curvature : Rate of change of slope angle of the curve with respect to distance along the curve



$$\rho \Delta\phi = \Delta s$$

$$\Rightarrow \frac{1}{\rho} = \lim_{\Delta s \rightarrow 0} \frac{\Delta\phi}{\Delta s}$$

$$\Rightarrow \frac{1}{\rho} = \frac{d\phi}{ds}$$

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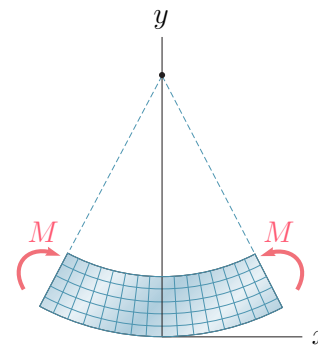
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Pure Bending

Deformations and Strains



Observations :

- top material fiber reduces in length (undergoes compression)
- bottom material fiber increases in length (undergoes tension)
- there is one material fiber in the plane of symmetry that does not change in length - **Neutral Axis**

Where is the location of the Neutral axis?

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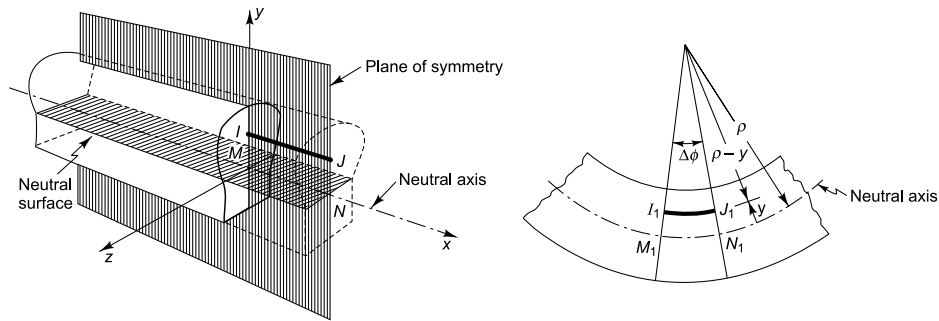
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Deformations and Strains



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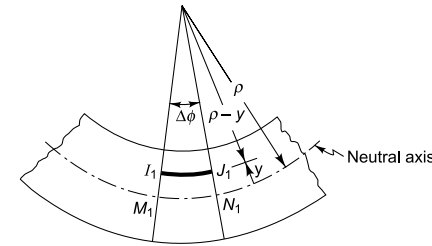
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Pure Bending

Deformations and Strains



- Length of neutral axis (N.A):
 $L = \rho \Delta \phi$ (Undeformed Line)

- Length of arbitrary line located at any distance y from the N.A is:
 $L' = (\rho - y) \Delta \phi$

- Change in length is:

$$\delta = L' - L = (\rho - y) \Delta \phi - \rho \Delta \phi = -y \Delta \phi$$

- Normal strain then is:

$$\epsilon_x = \frac{\delta}{L} = \frac{-y \Delta \phi}{\rho \Delta \phi} = -\frac{y}{\rho}$$

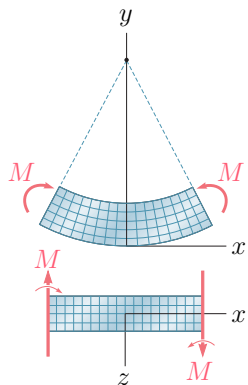
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Pure Bending

Deformations and Strains



Further Assumptions :

- deformations are sufficiently small - so undeformed coordinates provide an adequate approximation to location of points post deformation
- ϵ_x describes the longitudinal strain of all points in the cross-section
- Based on Symmetry :
 - $\gamma_{xy} = 0; \quad \gamma_{xz} = 0$
- ϵ_y, ϵ_z and γ_{yz} have to be symmetric about the plane of symmetry.

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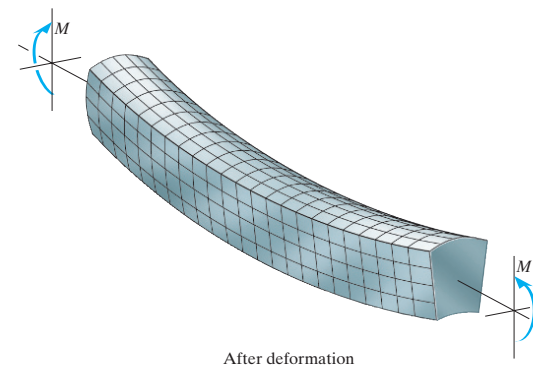
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Pure Bending

Hooke's Law - Stresses due to bending



For a linear elastic material :

$$\epsilon_x = \frac{1}{E} [\sigma_x - \nu(\sigma_y + \sigma_z)] = -\frac{y}{\rho}$$

$$\gamma_{xy} = \frac{\tau_{xy}}{G} = 0$$

$$\gamma_{xz} = \frac{\tau_{xz}}{G} = 0$$

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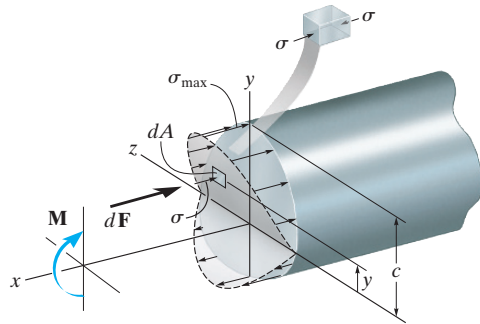
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Pure Bending

Static Equilibrium Conditions



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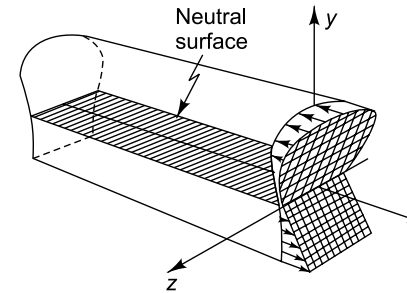
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13 / 16

Pure Bending

Static Equilibrium Conditions



- In case of pure bending, the resultant normal forces acting on the cross-sectional area must be zero.

$$\int_A \sigma_x dA \Rightarrow \int_A \left(\frac{-y}{\rho} E \right) dA = 0$$

$$\Rightarrow \int_A y dA = 0$$

- The neutral axis passes through the centroid of the cross-section (as long as stresses are in the elastic range)

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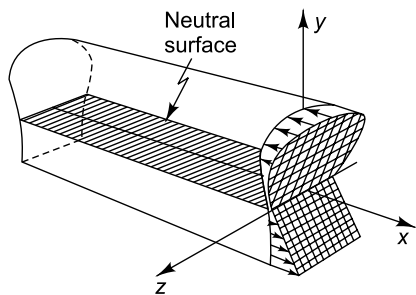
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14 / 16

Pure Bending

Static Equilibrium Conditions



- In case of pure bending, the moment of internal forces about neutral axis must be equal to M.

$$\int_A (-y) \sigma_x dA = \int_A (-y) \left(\frac{-y}{\rho} E \right) dA = M$$

$$\frac{E}{\rho} \int_A y^2 dA = \frac{E}{\rho} I_{zz} = M \Rightarrow \frac{1}{\rho} = \frac{M}{EI_{zz}}$$

$$\therefore \sigma_x = \frac{-y}{\rho} E = -\frac{My}{I_{zz}}$$

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15 / 16

Reading Assignment

Chapter 7 Crandall et. al.

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16 / 16